

Trainings ▾

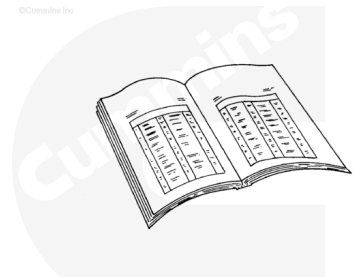
Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



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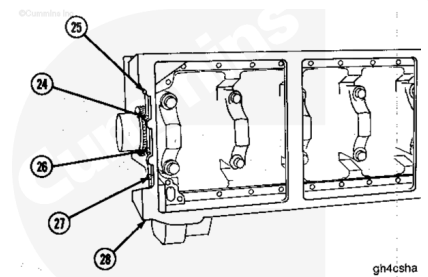
- Remove the accessory drive pulley. Refer to Procedure 009-004 (/qs3/pubsys2/xml/en/procedures/18/18-009-004-tr.html).
- Remove the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html).
- Drain lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html).
- Remove the engine barring device. Refer to Procedure 009-035 (/qs3/pubsys2/xml/en/procedures/18/18-009-035-tr.html).
- Remove the fuel pump. Refer to Procedure 005-016 (/qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html).
- Remove the air compressor. Refer to Procedure 012-014 (/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html).
- Remove the fuel pump drive. Refer to Procedure 009-011 (/qs3/pubsys2/xml/en/procedures/18/18-009-011-tr.html).
- Remove the hydraulic pump drive. Refer to Procedure 009-016 (/qs3/pubsys2/xml/en/procedures/18/18-009-016-

tr.html).

- Remove the gear cover spacer plate. Refer to Procedure 001-033 in Section 1.
(/qs3/pubsys2/xml/en/procedures/18/18-001-033-tr.html)
- Bar the engine, by turning the crankshaft, until the index marks on the camshaft gear and the crankshaft gear are in alignment.
- Remove the camshaft idler gear. Refer to Procedure 001-036 (/qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html).
- Remove the water pump idler gear. Refer to Procedure 001-040 (/qs3/pubsys2/xml/en/procedures/18/18-001-040-tr.html).
- Remove the hydraulic pump idler gear, if equipped. Refer to Procedure 001-039
(/qs3/pubsys2/xml/en/procedures/18/18-001-039-tr.html).
- Remove the camshaft gear or camshaft. Refer to Procedure 001-012
(/qs3/pubsys2/xml/en/procedures/18/18-001-012.html) or Refer to Procedure 001-008
(/qs3/pubsys2/xml/en/procedures/18/18-001-008-tr.html).
- Remove the oil pan or oil pan adapter cover. Refer to Procedure 007-025
(/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html). or Refer to Procedure 007-026
(/qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html).

Remove

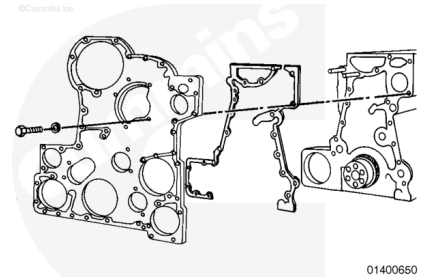
Remove the five capscrews that attach the bottom of the gear housing to the oil pan adapter.



Remove and tag the 11 gear housing capscrews.

Remove the gear housing and gasket.

Discard the gasket.



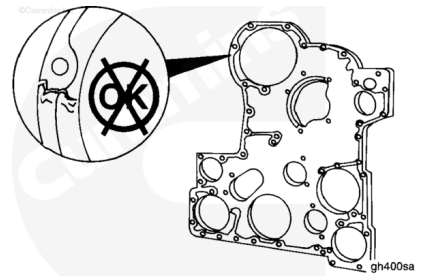
Clean and Inspect for Reuse

Clean the gear housing.

Check both sides of the gear housing gasket surfaces for fretting damage.

Inspect the gear housing for cracks.

If the gear housing is cracked or damaged it **must** be repaired or replaced.

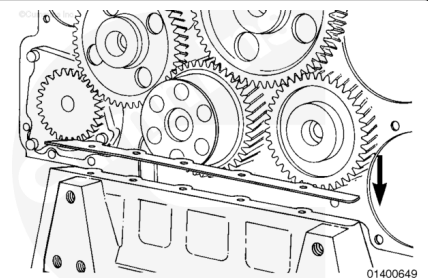


Install

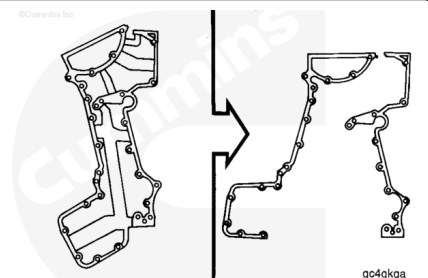
Install the gasket piece onto the top of the oil pan adapter.

Use a spray adhesive to hold it in position.

Apply sealant, Part Number 3164067, or equivalent, to the joints of the gasket.



Separate the gasket as illustrated in the graphic. Discard the center tab sections.



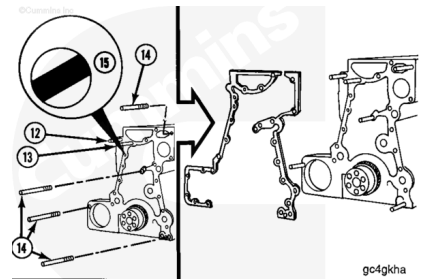
If the engine has a belt driven fan hub, install two studs (12) and (13).

The length of stud (12) is 140 mm [5½ in].

The length of stud (13) is 152 mm [6 in].

The diamond dowel (15) **must** be installed with the flat surface turned towards the master dowel hole at the lower right hand corner of the block.

Guide studs (14) will help during installation.



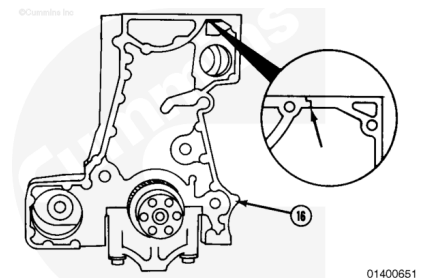
⚠ CAUTION ⚠

Do not use gasket cement. Damage to the gasket will result.

Note : The gasket illustrated in the graphic is used on old and new style blocks. When used on the old style block, there will be a capscrew hole (16) that is not used. It is not necessary to trim the gasket at this location.

Apply a small amount of sealant, Part Number 3164067, or equivalent, to both sides of the gasket at the butt joint.

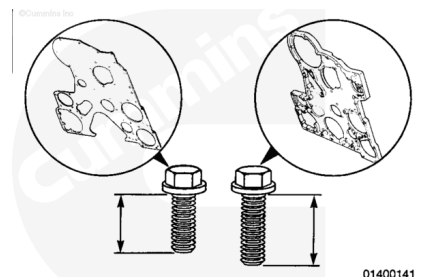
Install the gasket.



⚠ CAUTION ⚠

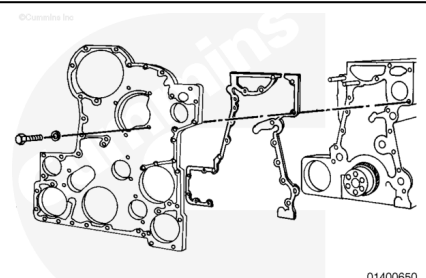
**Special capscrews are required to attach the gear housing.
The capscrews have a captive, cone shaped washer to maintain torque.**

The capscrews for the gear housing are 28.575 mm [1 1/8 in] long.



Install the gear housing.

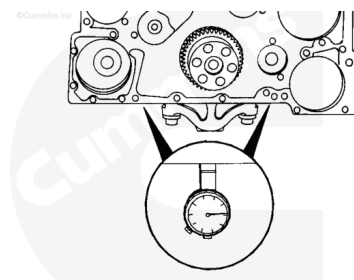
Install the capscrews and remove the guide studs.



Note : If the oil pan adapter is removed, the gear housing alignment must be checked. If the oil pan adapter is installed the alignment does not need to be checked.

Use depth gauge, Part Number 3164438, or equivalent to measure the distance from the bottom of the cylinder block to the bottom of the housing.

The bottom of the housing **must** be within 0.05 mm [0.002 in] of the bottom of the block.



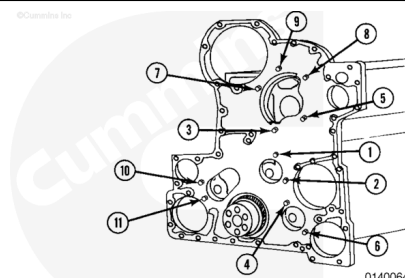
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Align the housing with the bottom of the block if necessary.

Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

Measure the housing to the block alignment to make sure it is within specifications.

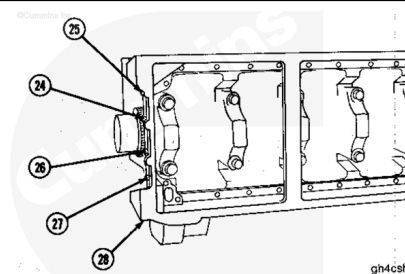


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Install the remaining 3/8 inch washers and capscrews into the bottom of the gear housing.

Tighten the capscrews.

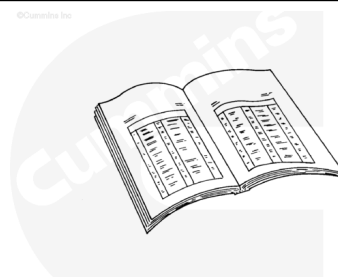
Torque Value: 45 n•m [33 ft-lb]



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Finishing Steps

- Install the oil pan or oil pan adapter cover. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>). or Refer to Procedure 007-026 (</qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html>).
- Install the camshaft gear or camshaft. Refer to Procedure 001-012 (</qs3/pubsys2/xml/en/procedures/18/18-001-012.html>) or Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/18/18-001-008-tr.html>).
- Install the hydraulic pump idler gear, if equipped. Refer to Procedure 001-039 (</qs3/pubsys2/xml/en/procedures/18/18-001-039-tr.html>).



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- Install the water pump idler gear. Refer to Procedure 001-040 (</qs3/pubsys2/xml/en/procedures/18/18-001-040-tr.html>).
- Install the camshaft idler gear. Refer to Procedure 001-036 (</qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html>).
- Install the gear cover spacer plate. Refer to Procedure 001-033 in Section 1.
(</qs3/pubsys2/xml/en/procedures/18/18-001-033-tr.html>)
- Bar the engine, by turning the crankshaft, until the index marks on the camshaft gear and the crankshaft gear are in alignment.
- Install the hydraulic pump drive. Refer to Procedure 009-016 (</qs3/pubsys2/xml/en/procedures/18/18-009-016-tr.html>).
- Install the fuel pump drive. Refer to Procedure 009-011 (</qs3/pubsys2/xml/en/procedures/18/18-009-011-tr.html>).
- Install the air compressor. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html>).
- Install the fuel pump. Refer to Procedure 005-016 (</qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html>).
- Install the engine barring device. Refer to Procedure 009-035 (</qs3/pubsys2/xml/en/procedures/18/18-009-035-tr.html>).
- Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>).
- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/18/18-001-031-tr.html>).
- Install the accessory drive pulley. Refer to Procedure 009-004 (</qs3/pubsys2/xml/en/procedures/18/18-009-004-tr.html>).

